

Proximal Operators and the Proximal Gradient Method: A Reference Table

1 Setting and basic definitions

We consider the composite problem

$$\min_{x \in \mathbb{R}^n} F(x) := f(x) + g(x), \quad (1)$$

where f is convex and differentiable with L -Lipschitz gradient, $\|\nabla f(x) - \nabla f(y)\|_2 \leq L \|x - y\|_2$, and g is proper, closed, and convex (possibly nonsmooth). Norms are Euclidean unless indicated.

Definition 1 (Proximal operator). For $t > 0$ and $v \in \mathbb{R}^n$,

$$\text{prox}_{tg}(v) := \arg \min_{z \in \mathbb{R}^n} \left\{ g(z) + \frac{1}{2t} \|z - v\|_2^2 \right\}. \quad (2)$$

For proper, closed, convex g the objective in (2) is strongly convex, so the minimizer exists and is unique.

Definition 2 (Auxiliary notation).

- Indicator of a set C : $\iota_C(x) = 0$ if $x \in C$ and $+\infty$ otherwise.
- Euclidean projection onto a closed convex set C : $\Pi_C(v) := \arg \min_{z \in C} \|z - v\|_2$. Hence $\text{prox}_{t\iota_C}(v) = \Pi_C(v)$ for every $t > 0$.
- Soft thresholding with level $\tau \geq 0$, applied entrywise: $[\mathcal{S}_\tau(v)]_i := \text{sign}(v_i) \max\{|v_i| - \tau, 0\}$.
- Positive part $(a)_+ := \max\{a, 0\}$ (entrywise for vectors).
- Subdifferential $\partial g(x)$; convex conjugate $g^*(u) := \sup_x \{\langle u, x \rangle - g(x)\}$.

Proposition 1 (Optimality condition for the prox). *If g is proper, closed, and convex, then*

$$z = \text{prox}_{tg}(v) \iff \frac{v - z}{t} \in \partial g(z).$$

Proof. The objective in (2) is convex, so z minimizes it iff $0 \in \partial g(z) + \frac{1}{t}(z - v)$. □

Proposition 2 (Moreau decomposition). *For proper, closed, convex g and $t > 0$,*

$$v = \text{prox}_{tg}(v) + t \text{prox}_{t^{-1}g^*}(v/t) \quad \text{for all } v \in \mathbb{R}^n.$$

2 The proximal gradient update

Proposition 3 (Proximal gradient step). *For any $x \in \mathbb{R}^n$, the following three definitions of x^+ coincide:*

$$x^+ = \arg \min_z \left\{ \frac{L}{2} \left\| z - \left(x - \frac{1}{L} \nabla f(x) \right) \right\|_2^2 + g(z) \right\} \quad (3)$$

$$= \text{prox}_{g/L} \left(x - \frac{1}{L} \nabla f(x) \right) \quad (4)$$

$$= \arg \min_z \left\{ f(x) + \langle \nabla f(x), z - x \rangle + \frac{L}{2} \|z - x\|_2^2 + g(z) \right\}. \quad (5)$$

Proof. (3)=(4): with $t = 1/L$ in (2), $\frac{1}{2t} \|z - v\|_2^2 = \frac{L}{2} \|z - v\|_2^2$, and take $v = x - \frac{1}{L} \nabla f(x)$.

(3)=(5): expanding the square,

$$\frac{L}{2} \left\| (z - x) + \frac{1}{L} \nabla f(x) \right\|_2^2 = \frac{L}{2} \|z - x\|_2^2 + \langle \nabla f(x), z - x \rangle + \frac{1}{2L} \|\nabla f(x)\|_2^2.$$

The objectives in (3) and (5) therefore differ by the constant $f(x) - \frac{1}{2L} \|\nabla f(x)\|_2^2$, which does not depend on z , so they have the same minimizer. $\square$

Remark. Form (5) shows that x^+ minimizes g plus a quadratic upper model of f at x (valid because ∇f is L -Lipschitz). With a step size $t \in (0, 1/L]$ the step is $x^+ = \text{prox}_{tg}(x - t\nabla f(x))$. In the tables below, set $t = 1/L$ and

$$v := x - \frac{1}{L} \nabla f(x)$$

to read off $x^+ = \text{prox}_{tg}(v)$ directly.

Proposition 4 (Fixed points). x^* minimizes F in (1) if and only if $x^* = \text{prox}_{g/L}(x^* - \frac{1}{L} \nabla f(x^*))$.

Proof. By Proposition 1 with $t = 1/L$ and $v = x^* - \frac{1}{L} \nabla f(x^*)$, the fixed-point equation holds iff $L(v - x^*) = -\nabla f(x^*) \in \partial g(x^*)$, i.e. $0 \in \nabla f(x^*) + \partial g(x^*)$, which is the optimality condition for the convex problem (1). $\square$

3 Proximal operators: closed forms

In every table, $t > 0$ is the prox parameter (use $t = 1/L$ for the proximal gradient step) and $\lambda, \lambda_1, \lambda_2 \geq 0$.

Table 1: Vector regularizers. Rows 3–7 act entrywise; row 9 acts groupwise.

$g(x)$	$\text{prox}_{tg}(v)$	Remarks
1 0	v	Plain gradient step.
2 $\langle c, x \rangle$	$v - tc$	Linear term.
3 $\lambda \ x\ _1$	$\mathcal{S}_{\lambda t}(v)$	Lasso ; soft thresholding at level λt ($= \lambda/L$).
4 $\lambda \sum_i w_i x_i , w_i \geq 0$	$\left[\text{sign}(v_i) (v_i - \lambda t w_i)_+ \right]_{i=1}^n$	Weighted / adaptive Lasso.
5 $\frac{\lambda}{2} \ x\ _2^2$	$\frac{v}{1 + \lambda t}$	Ridge (Tikhonov).
6 $\lambda_1 \ x\ _1 + \frac{\lambda_2}{2} \ x\ _2^2$	$\frac{\mathcal{S}_{\lambda_1 t}(v)}{1 + \lambda_2 t}$	Elastic net: threshold first, then shrink.
7 $\lambda \ x\ _1 + \iota_{\{x \geq 0\}}(x)$	$(v - \lambda t)_+$	Nonnegative Lasso.
8 $\lambda \ x\ _2$	$\left(1 - \frac{\lambda t}{\ v\ _2} \right)_+ v$	Block soft thresholding; equals 0 if $\ v\ _2 \leq \lambda t$ (incl. $v = 0$).
9 $\lambda \sum_{G \in \mathcal{G}} w_G \ x_G\ _2$	$\left[\left(1 - \frac{\lambda t w_G}{\ v_G\ _2} \right)_+ v_G \right]_{G \in \mathcal{G}}$	Group Lasso; groups $\mathcal{G}$ must <i>not</i> overlap.
10 $\lambda \ x\ _\infty$	$v - \Pi_{\{\ u\ _1 \leq \lambda t\}}(v)$	Via Moreau (Prop. 2); ℓ_1 -ball projection in Table 2.

Table 2: Indicator functions $g = \iota_C$. Here $\text{prox}_{tg} = \Pi_C$ does not depend on t .

C	$\Pi_C(v)$	Remarks
11 $\{x \geq 0\}$	$(v)_+$	Entrywise.
12 $\{\ell \leq x \leq u\}$	$\min\{\max\{v, \ell\}, u\}$	Box; entrywise clipping.
13 $\{\ x\ _2 \leq r\}$	$\min\left\{1, \frac{r}{\ v\ _2}\right\} v$	ℓ_2 ball (v unchanged if inside).
14 $\{a^\top x \leq \beta\}, a \neq 0$	$v - \frac{(a^\top v - \beta)_+}{\ a\ _2^2} a$	Halfspace.
15 $\{Ax = b\}, A$ full row rank	$v - A^\top (AA^\top)^{-1}(Av - b)$	Affine set.
16 $\Delta_r = \{x \geq 0, \mathbf{1}^\top x = r\}$	$(v - \theta \mathbf{1})_+$	Simplex, $r > 0$; θ from Eq. (6).
17 $\{\ x\ _1 \leq r\}$	$\begin{cases} v, & \ v\ _1 \leq r \\ \text{sign}(v) \odot \Pi_{\Delta_r}(v), & \text{otherwise} \end{cases}$	ℓ_1 ball; $\odot$ entrywise product.

Simplex threshold θ . Sort v in decreasing order, $u_1 \geq u_2 \geq \dots \geq u_n$, and set

$$\rho = \max \left\{ j \in \{1, \dots, n\} : u_j - \frac{1}{j} \left(\sum_{i=1}^j u_i - r \right) > 0 \right\}, \quad \theta = \frac{1}{\rho} \left(\sum_{i=1}^{\rho} u_i - r \right). \quad (6)$$

The cost is $O(n \log n)$.

 Table 3: Matrix functions ($V \in \mathbb{R}^{m \times n}$ with SVD $V = U \text{diag}(\sigma) W^\top$) and nonconvex penalties.

g	$\text{prox}_{tg}(\cdot)$	Remarks
18 $\lambda \ X\ _*$	$U \text{diag}((\sigma - \lambda t)_+) W^\top$	Singular value thresholding.
19 $\iota_{\{X \succeq 0\}}$	$Q \text{diag}((\mu)_+) Q^\top$	For symmetric $V = Q \text{diag}(\mu) Q^\top$ (symmetrize a general V first).
<i>Nonconvex: the prox exists but may be set-valued. Props. 1, 2, 4 require convexity and do not apply; Prop. 3 still holds.</i>		
20 $\lambda \ x\ _0$	$[z_i]_{i=1}^n, z_i = \begin{cases} v_i, & v_i > \sqrt{2\lambda t} \\ 0, & v_i < \sqrt{2\lambda t} \end{cases}$	Hard thresholding; at $ v_i = \sqrt{2\lambda t}$ both 0 and v_i are minimizers.
21 $\iota_{\{\ x\ _0 \leq s\}}$	keep the s largest $ v_i $, zero the rest	Non-unique under ties.
22 $\iota_{\{\text{rank}(X) \leq k\}}$	$U \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0) W^\top$	Truncated SVD (Eckart–Young); non-unique if $\sigma_k = \sigma_{k+1}$.

 Table 4: Calculus rules (h proper, closed, convex; $t, \rho > 0$; $\alpha \neq 0$). These generate new rows from known ones.

$g(x)$	$\text{prox}_{tg}(v)$	Remarks
$\sum_{i=1}^n h_i(x_i)$	$[\text{prox}_{th_i}(v_i)]_{i=1}^n$	Separable sum.
$h(x) + \frac{\rho}{2} \ x\ _2^2$	$\text{prox}_{\frac{t}{1+\rho t}h} \left(\frac{v}{1+\rho t} \right)$	Gives row 6 from row 3.
$h(x) + \langle c, x \rangle$	$\text{prox}_{th}(v - tc)$	
$h(x - a)$	$a + \text{prox}_{th}(v - a)$	Translation.
$h(\alpha x)$	$\frac{1}{\alpha} \text{prox}_{\alpha^2 th}(\alpha v)$	Scaling.

4 Derivation of soft thresholding (row 3)

Row 3 is separable, so it suffices to solve the scalar problem with $\tau = \lambda t$:

$$z^* = \arg \min_{z \in \mathbb{R}} \tau |z| + \frac{1}{2}(z - v)^2.$$

The optimality condition is $0 \in z - v + \tau \partial|z|$, where $\partial|z| = \{\text{sign}(z)\}$ for $z \neq 0$ and $\partial|0| = [-1, 1]$.

- If $z^* > 0$: $z^* = v - \tau$, which is consistent iff $v > \tau$.
- If $z^* < 0$: $z^* = v + \tau$, which is consistent iff $v < -\tau$.
- If $z^* = 0$: we need $v \in \tau[-1, 1]$, i.e. $|v| \leq \tau$.

The three cases partition $\mathbb{R}$, so $z^* = \text{sign}(v) \max\{|v| - \tau, 0\} = \mathcal{S}_\tau(v)$. The prox objective $\lambda|z| + \frac{1}{2t}(z - v)^2$ from (2) equals $\frac{1}{t}$ times the objective above, so the two share the minimizer, which gives $\text{prox}_{t\lambda\|\cdot\|_1}(v) = \mathcal{S}_{\lambda t}(v)$ entrywise.

5 Example: the Lasso

For $\min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|x\|_1$ we have

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2, \quad \nabla f(x) = A^\top (Ax - b), \quad L = \lambda_{\max}(A^\top A) = \|A\|_2^2, \quad g(x) = \lambda \|x\|_1.$$

By Proposition 3 and row 3 of Table 1, the proximal gradient update is

$$x^{k+1} = \mathcal{S}_{\lambda/L} \left(x^k - \frac{1}{L} A^\top (Ax^k - b) \right), \quad \text{i.e.} \quad x_i^{k+1} = \text{sign}(v_i) \max \left\{ |v_i| - \frac{\lambda}{L}, 0 \right\}, \quad v = x^k - \frac{1}{L} A^\top (Ax^k - b). \quad (7)$$

Algorithm 1 ISTA and FISTA for the Lasso

Require: $A, b, \lambda; L = \|A\|_2^2; x^0$ (e.g. 0); set $y^1 = x^0, t_1 = 1$

- 1: **for** $k = 1, 2, \dots$ **do**
 - 2: **ISTA:** $x^k = \mathcal{S}_{\lambda/L} \left(x^{k-1} - \frac{1}{L} A^\top (Ax^{k-1} - b) \right)$
 - 3: **FISTA:** $x^k = \mathcal{S}_{\lambda/L} \left(y^k - \frac{1}{L} A^\top (Ay^k - b) \right)$
 - 4: $t_{k+1} = \frac{1}{2} (1 + \sqrt{1 + 4t_k^2}), \quad y^{k+1} = x^k + \frac{t_k - 1}{t_{k+1}} (x^k - x^{k-1})$
 - 5: **end for**
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Convergence (Beck–Teboulle, 2009). With constant step $1/L$ and any minimizer x^* ,

$$\text{ISTA: } F(x^k) - F^* \leq \frac{L \|x^0 - x^*\|_2^2}{2k}, \quad \text{FISTA: } F(x^k) - F^* \leq \frac{2L \|x^0 - x^*\|_2^2}{(k+1)^2}.$$

ISTA is monotone, $F(x^{k+1}) \leq F(x^k)$; FISTA in general is not.

Practical notes.

- *Stopping.* By Proposition 4, x^* is optimal iff it is a fixed point of the update, so a natural criterion is the gradient-mapping norm $\|L(x^k - x^{k+1})\|_2 \leq \varepsilon$.
- *Largest useful λ .* $x^* = 0$ is optimal iff $\lambda \geq \|A^\top b\|_\infty$ (since $0 \in -A^\top b + \lambda[-1, 1]^n$ exactly when $\|A^\top b\|_\infty \leq \lambda$). Regularization paths usually start from $\lambda_{\max} = \|A^\top b\|_\infty$.
- *Unknown L .* Use backtracking: increase L until $f(x^+) \leq f(x) + \langle \nabla f(x), x^+ - x \rangle + \frac{L}{2} \|x^+ - x\|_2^2$ holds, which is the condition that makes the model in (5) an upper bound at x^+ .

Verification. Every row of Tables 1–4, Propositions 2–4, the Lasso threshold $\lambda_{\max}$, and the ISTA/FISTA rate bounds were checked numerically against an independent convex solver (cvxpy/Clarabel) on random instances; nonconvex rows were checked by exhaustive search or the Eckart–Young theorem.